

Public Perception and Misinformation in GMOs – Case Study (2019–2021)

1. Background & Importance

- **Misinformation:** Information that conflicts with widely accepted scientific consensus.
 - **Problematic Areas:** Climate change, vaccines, nuclear power, GMOs (genetically modified organisms).
 - **GMOs:** Decades of misinformation campaigns have led to negative public perception and restrictive policies globally.
 - **Scientific Consensus:** GMOs are considered safe by major scientific organizations (e.g., US National Academy of Sciences, AAAS, WHO, UK Royal Society).
-

2. Study Aim

- **Objective:** Quantify misinformation in English-language mainstream and online news media (2019–2021) about GMOs.
 - **Focus Areas:** Volume, readership, topics, sentiment, and regional patterns.
-

3. Methods

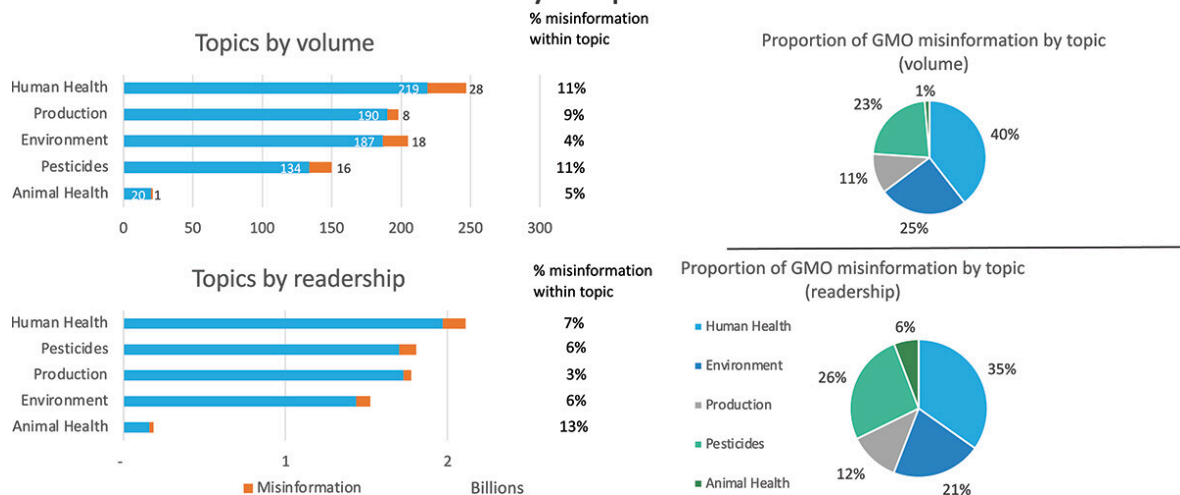
- **Database:** 535 articles from global media, including anglophone sub-Saharan African outlets.
 - **Article Categorization:** Five main topics:
 - **Human Health:** GMOs and human consumption.
 - **Environment:** GMOs and ecological impact.
 - **Production:** Yields, food security, crop control.
 - **Pesticides:** GMOs and herbicide/insecticide use.
 - **Animal Health:** GMOs in animal feed.
 - **Sentiment Analysis:** Articles coded as positive, negative, neutral, or mixed.
 - **Misinformation Definition:** Content deviating from the widely accepted scientific consensus.
-

4. Key Findings

● Volume & Misinformation Rates:

- 535 articles reviewed → 47 contained misinformation (~9%).
488 factual articles → 91%.
- Potential readership: 4.8B for factual articles, 256M for misinformation (~5% of total audience exposed to misinformation).

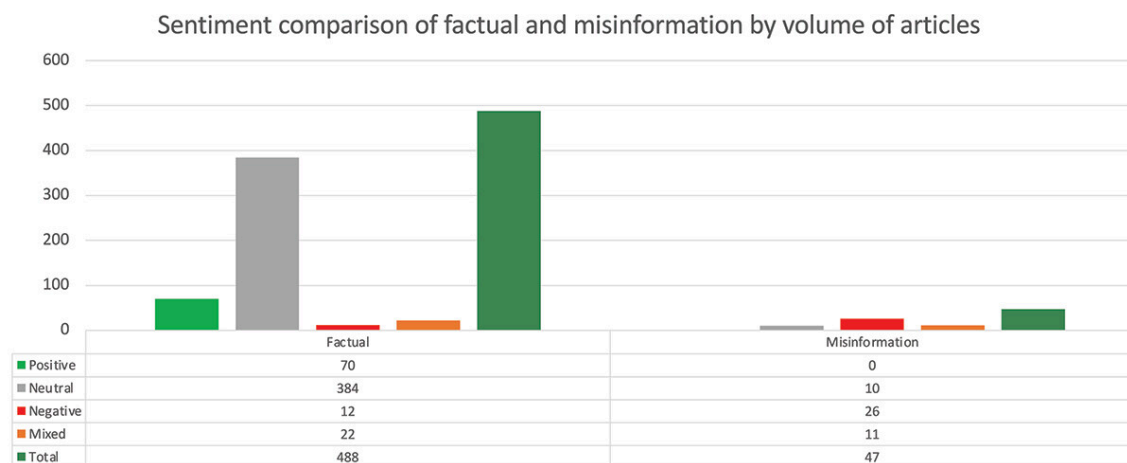
GMO misinformation by topic



● Misinformation by Topic:

- **Human Health:** 28/247 articles (11%).
- Pesticides:** 16/150 articles (11%).
- Environment:** ~25% of misinformation.
- Production:** 47M readership.
- Animal Health:** 1/21 articles (13% - highest % by readership).

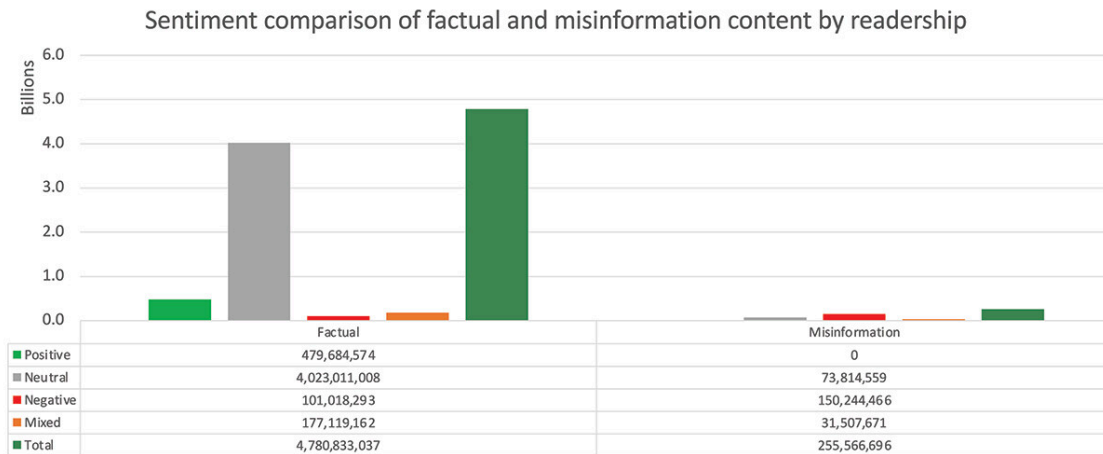
Sentiment overview - volume



- **Sentiment of Misinformation:**

- 100% of misinformation articles were negative, mixed, or neutral.
No positive misinformation detected, suggesting anti-GMO activism drives misinformation.

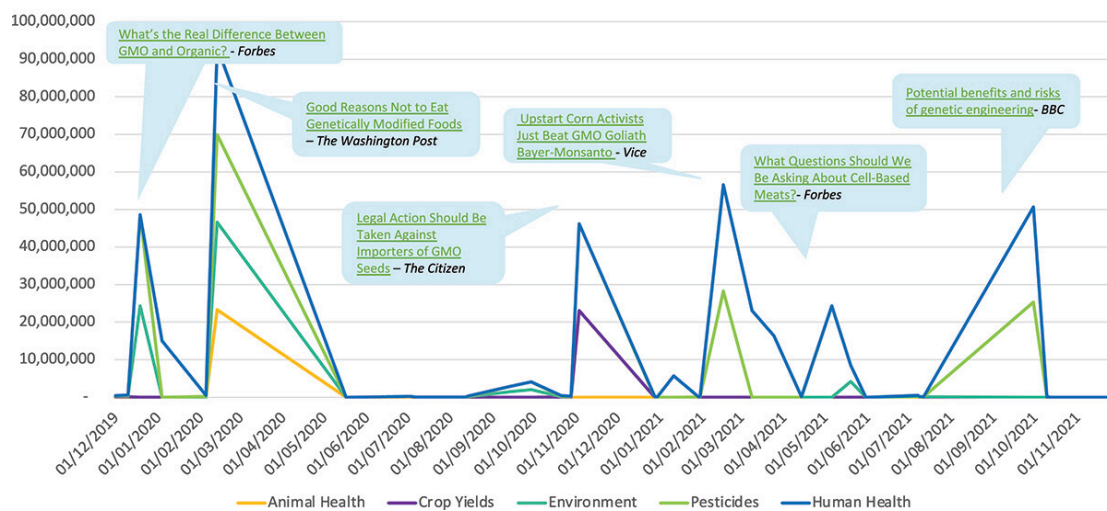
Sentiment overview - readership



- **Regional Patterns:**

- **Africa:** 20% of GMO articles = misinformation (highest proportion).
North America: 5%, **Europe:** 7%.
Bias: African media were deliberately overrepresented in the sample for special analysis.

Readership of GMO misinformation by topic



5. Impacts of Misinformation

- **Consequences:**

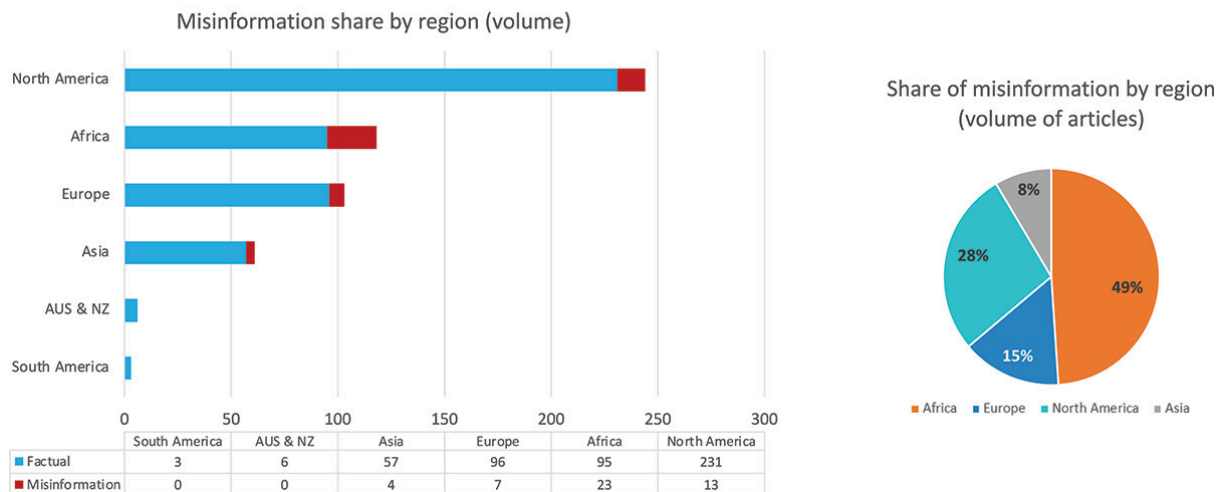
- Lower adoption of GMOs → lost yield gains, higher pesticide use.
Europe: Non-adoption → 33M tonnes CO₂ equivalent extra emissions.

Global Impact: 775.4M kg less pesticide use, ~15.3M cars worth of greenhouse gas reduction, \$18.9B farm-level net benefits.

Developing Countries:

- **Bangladesh:** Bt brinjal → higher profit, reduced pesticide use.
- **India:** Bt cotton → higher incomes, fewer poisonings.
- **Africa:** Misinformation slows adoption of GM cowpea → malnutrition risk, thousands of potential lives lost.

GMO misinformation in different regions



6. Causes of Misinformation

- **Activist Networks:** Anti-GMO networks, often based in the Global North, influencing African media.
"False Balance": Media presenting self-proclaimed experts who deny the scientific consensus.
Misinformation Tactics: Use of scientific-sounding claims to create doubt.

7. Conclusions & Recommendations

- **Misinformation Problem:** GMO misinformation is a significant issue in the media, surpassing even COVID-19 vaccine misinformation.
Impact on Public and Policy: Almost all misinformation is negative, shaping public and policy opposition.

Recommendations:

- The scientific community needs to improve communication with the media and the public.
Urgent focus on **sub-Saharan Africa** to ensure food security and informed policymaking.
<https://www.tandfonline.com/doi/full/10.1080/21645698.2022.2140568#d1e336>
-

8. Key Stats

- **Misinformation rate:** 9% of articles (47/535).
Readership of misinformation: 256M.
Sentiment: 100% non-positive.
Africa misinformation rate: 20% of GMO coverage.
Connection: Misinformation → Public Perception → Policy Impact → Real-world Consequences.
-

Case Study: An Unprecedented Outreach Event in Argentina on Gene Editing (2018)

Brief Description of the Research Project

- **Event Objective:** The event aimed to engage the Argentine public on the topic of gene editing (GE), particularly CRISPR technology, and its societal implications. The symposium was the first of its kind in Argentina, co-organized by multiple government ministries and agencies, including the **Ministry of Science, Health, and Agroindustry**.
- **Key Focus Areas:** Health research, food technology, and agriculture were emphasized, given their relevance to GE advancements.
- **Audience:** The event attracted **452 attendees** in-person and **196 live-stream viewers**, with **40,000+ impressions** on Twitter.
- **Media Coverage:** The event garnered significant mainstream media attention, with **19 news stories** published in major national newspapers shortly after.

Background

- **Argentina's Expertise in Biotechnology:** Argentina has a strong foundation in molecular biology and biotechnology, which has facilitated its advancement in GE. In 2018, the country declared its support for GE to the **World Trade Organization**.

- **Regulatory Framework:** Argentina's existing regulatory framework supports GE developments and research, helping to foster an environment for technological advancement.
- **Challenges:** The organizers were tasked with delivering an event without prior precedent or established protocols for communicating GE to the public.

Ethical Issues and Communication Challenges

1. Counteracting Prejudices:

- There was a need to overcome public prejudices against biotechnology, particularly regarding GMOs and transgenesis.
- **Strategy:** Simple language was used to explain the benefits of GE and address concerns, without making exaggerated promises or comparisons to GMOs.

2. Unified Messaging:

- The event required careful coordination to ensure all participants (government entities, speakers, and media) communicated a unified message.
- **Speaker Preparation:** To avoid misinformation, speakers were provided with training and clear guidelines. Ensuring everyone was on the same page was essential for presenting a consistent narrative.

3. Engaging the Audience:

- The public was invited to ask questions, covering topics like the safety of GE-based therapies and when they might become available.
- **Key Response:** The organizers were committed to providing truthful, transparent answers without making promises. They reassured the audience that GE developments were subject to rigorous safety assessments by regulatory bodies.

4. Post-Event Strategy:

- Although the event was successful, the organizers identified the need for a **post-event communication strategy** to maintain engagement and address any new questions or concerns from the public.
- **Key Learning:** Building trust through continuous, transparent communication is essential for maintaining public confidence in emerging

technologies.

Conclusions and Recommendations

1. Communication Tailoring:

- Communication methods should be specifically designed for the target audience. Clear objectives, focused content, and appropriate delivery channels are crucial.

2. Managing Expectations:

- **Avoid Over-Promising:** While CRISPR technology holds great potential, it should not be presented as a “magic bullet” for all societal problems. Unrealistic expectations can undermine trust in the technology.

3. Truthful Communication:

- When comparing CRISPR to other genetic technologies, it’s essential to avoid overselling one technique over others. Each technology has its own unique benefits and limitations, and these should be communicated honestly.

4. Broad Stakeholder Involvement:

- The success of the event was partly due to the collective effort of multiple stakeholders, including governmental bodies, researchers, and the media. Broader participation ensures a more credible and impactful message.

5. Ongoing Engagement:

- After the event, continuing the conversation through social media, follow-up discussions, and educational outreach is essential to keeping the public informed and engaged.

Impact of the Event

- **Positive Public Feedback:** The event received positive comments from attendees, and the Ministry of Science congratulated the team for its success.
- **Media Attention:** Beyond the event, two speakers were invited to speak on radio shows about gene editing, highlighting the long-term impact of effective science communication.

<https://www.gfbr.global/wp-content/uploads/2019/12/T4.-Sebastian-Barbosa-.pdf>

Key Takeaways for Science Communication

1. **Pre-event planning** and stakeholder alignment are critical for ensuring a unified, clear message.
 2. **Tailor your communication** to the target audience's level of understanding and concerns.
 3. **Transparency and honesty** are essential, especially when discussing emerging technologies that might raise public apprehension.
 4. A **post-event communication strategy** can help maintain momentum and address ongoing concerns.
 5. **Broad collaboration** across sectors, including media, can amplify the reach and credibility of science communication efforts.
-

A critical examination of how public policy related to autism has been shaped by misinformation, particularly in Europe, where public policy often stands in contrast to evidence-based practices like Applied Behaviour Analysis (ABA). It highlights the significant gap in autism interventions between North America and Europe, mainly due to the political and social influence of misinformation or lack of proper knowledge in the field of ABA.

Key Points:

- ❖ **Autism's Growing Impact:** Autism affects many worldwide, with high social and economic costs. Yet, many countries fail to use effective, evidence-based interventions.
- ❖ **ABA as the Gold Standard:** In North America, ABA (Applied Behaviour Analysis) is widely supported by research and policy. In Europe, however, ABA faces resistance, especially in the UK and Ireland.
- ❖ **Misinformation Shaping Policy:** Propaganda and misinformation influence public policy. For example, in the UK, the NICE review ignored ABA experts, leading to policies that dismiss its effectiveness.
 - (The **NICE review** (conducted by the UK's National Institute for Health and Care Excellence) **ignored ABA experts** means that when NICE was evaluating autism treatments, they did not include professionals who are specially trained in **Applied Behaviour Analysis (ABA)**, despite the fact that ABA is a well-researched and widely recognized method for supporting individuals with autism. As a result, the review made recommendations about autism interventions **without considering the expertise and evidence from ABA professionals**. This led to the **misrepresentation or undervaluation of ABA**, and in some cases, it caused **policies that failed**

to support or recommend ABA-based treatments, even though there's extensive evidence backing its effectiveness.

- ❖ **Political Bias and Propaganda:** Reports and reviews are often swayed by political agendas, using tactics like ignoring evidence and attacking critics, keeping ABA from gaining acceptance.
- ❖ **Northern Ireland Case Study:** In Northern Ireland, government reports on autism ignored ABA experts, spreading misinformation and leaving families without effective support.
- ❖ **Lack of ABA Training in Europe:** Many European countries lack trained ABA professionals, leading to flawed autism treatment guidelines and poor policy decisions.
- ❖ **Ethical Failures:** Governments have excluded ABA experts from key panels, ignoring research that contradicts their policies, resulting in ineffective autism interventions.

Conclusion: Misinformation harms autism policy. The paper calls for evidence-based decision-making and greater inclusion of ABA experts to ensure better interventions for individuals with autism.

The current status and regulation of transgenic (genetically modified) crops in India, specifically focusing on the ongoing debate around the testing and approval of GM cotton seeds, such as those containing the Cry2Ai gene, which is designed to make cotton resistant to the pink bollworm pest.

1. **Transgenic Crops in India:**

- **Cotton is the only transgenic crop currently cultivated commercially in India**, although other crops like brinjal, tomato, and maize are undergoing trials.
- **The Genetic Engineering Appraisal Committee (GEAC)** is responsible for evaluating GM crop proposals. Recently, they approved the environmental release of the GM mustard hybrid DMH-11 for seed production and testing, though the full commercial approval is still pending.

2. **Regulatory Process:**

- The approval process for GM crops in India is complex, involving multiple safety assessments and trials at agricultural universities and research centers. These trials assess the environmental impact and the crop's

performance compared to non-GM varieties.

- Transgenic crops must prove their superiority in specific traits (e.g., pest resistance or drought tolerance) and ensure they don't harm surrounding ecosystems.

3. **Opposition from States:**

- **Three states — Gujarat, Maharashtra, and Telangana** — have deferred/postponed the proposal to test GM cotton with the Cry2Ai gene, which is meant to combat pink bollworm.
- While Haryana approved the testing, these other states cited concerns about the impact and necessity of GM crops in their regions.
- Some activist groups are pushing for stricter regulation and caution, arguing that GM crops may have unintended ecological consequences.

4. **Legal and Political Landscape:**

- There is ongoing litigation in the Supreme Court regarding the permissibility of GM food crops in India, with petitioners raising concerns about their environmental impact and the use of banned herbicides.
- A significant hurdle is that agriculture is a state subject, meaning individual states have considerable autonomy in approving or rejecting the trials.

5. **Proposed Changes:**

- To address delays in state-level approvals, the GEAC is considering designating specific sites across India as "notified testing sites," where GM crop trials could be conducted without needing state permission.
- This would speed up the process but is controversial, as it could bypass state objections.

Key Players:

- **GEAC:** The central body overseeing the regulation of GM crops.
- **DBT (Department of Biotechnology):** A government department involved in GM crop approvals.
- **ICAR (Indian Council of Agricultural Research):** Conducts research and field trials for GM crops.

- **State Governments:** Have significant authority over the approval of field trials within their regions.
- **Activist Groups:** Raise concerns about the safety and environmental impact of GM crops.

This issue highlights the tension between technological advancements in agriculture and the concerns about their long-term ecological and health impacts, especially when the regulatory framework is still evolving.



India's ongoing struggle with regulating genetically modified (GM) crops, particularly in light of a significant Supreme Court directive in March 2025 that urges the government to formulate a comprehensive national policy. It touches on the challenges of balancing scientific advancement with public trust, political dynamics, and the future of agricultural policy in the country. Here's a breakdown of the key points:

1. Political Uncertainty and Public Mistrust:

- For over two decades, India's GM crop regulation has been caught in a web of political uncertainty and public skepticism. **Despite being an early adopter of transgenic crops like Bt cotton, the country has struggled to extend the success of GM crops to food crops like GM mustard.**
- India has a strong research community developing transgenic crops (e.g., **drought-resistant rice, pest-resistant wheat**), but the regulatory bottleneck has led to stagnation. This stagnation is not due to a lack of scientific capability but rather the governance **system's inability to move forward due to public protests**, ideological opposition, and political caution.

2. Regulatory Challenges:

- The **Genetic Engineering Appraisal Committee (GEAC)**, responsible for regulating GM crops, faces pressure from both evidence and ideology. Regulatory decisions are often **influenced more by public perception and protests than by scientific data**, which has created an innovation–acceptance gap.
- This has resulted in a **lack of a clear, transparent route for approval**, leading to waning private-sector investment, diminished public-sector morale, and, most importantly, a missed opportunity for farmers who could benefit from the technology.
<https://www.nature.com/articles/d44151-025-00098-y>

3. The Role of Public Mistrust:

- Public distrust towards GM crops in India is largely driven by a combination of **misinformation, lack of transparency, and polarized discourse**. Campaigns emphasizing “seed sovereignty” and the dangers of “toxic crops” have shaped public opinion, often more than scientific evidence.
- The regulatory bodies have not been proactive in communicating risk assessments, field trials, or biosafety reviews in an accessible manner, leading to a gap between scientific facts and public perception.

4. The Call for Transparency:

- A **transparent GM policy** is seen as crucial for overcoming public skepticism. This transparency would include making all safety assessments, trial results, and regulatory decisions publicly available in formats accessible to both experts and the general public.
- The article stresses that only through such transparency can the public’s reaction shift from **reactive opposition** to **informed engagement** with the technology.

5. Global Lessons and Local Context:

- The article points to the experiences of **the United States, Brazil, and China**—countries that have successfully integrated GM crops into their agricultural systems with **robust regulatory frameworks** and **farmer-led adoption**.
- While India need not mimic these models exactly, there is clear global evidence that with proper safeguards, **GM crops can coexist** with ecological stewardship, food safety, and farmer autonomy. Ignoring these technologies could lead to rising import costs or missed opportunities for agricultural productivity.

6. Principles for a Forward-Looking GM Policy:

The article proposes six guiding principles for a forward-thinking GM policy:

- **Independent, evidence-based regulation:** A depoliticized regulatory system focused on biosafety science.
- **Transparency and open data:** Full public access to safety, efficacy, and trial results.
- **Structured public engagement:** Inclusion of farmers, scientists, civil society, and industry in consultation processes.
- **Post-approval oversight:** Environmental monitoring, labeling, and liability systems for unforeseen consequences.
- **Public investment in research:** Increased funding for regional crop traits, especially from publicly funded universities and research institutes.
- **Choice and coexistence:** A policy that allows farmers to choose between GM and traditional methods, with safeguards for organic farming.

7. The Need for a New Green Consensus:

- The article argues that India can craft a **unique national GM policy** rooted in scientific excellence and democratic engagement, ensuring **inclusive agricultural growth**. This would not be about **corporate control** over agriculture, but about leveraging Indian scientific capabilities to meet the nation's agricultural and food security needs.
- The comparison to the **Green Revolution** is made, suggesting that, just as that revolution succeeded because of a national consensus around science-led agriculture, India needs a similar consensus in the era of climate change and agricultural adaptation.

Conclusion:

The piece emphasizes that India has the scientific potential to become a leader in GM crop adoption, but **clear governance** and **public engagement** are critical to overcoming the current regulatory deadlock. The challenge is not just scientific, but also political and societal. The 2025 Supreme Court directive could serve as a pivotal moment for the country to rethink its approach to GM crops, balancing innovation with transparency and public trust.
